

Generate Sigma rules for: • Windows Event ID 4625 • Sysmon Event ID 1 • Sysmon Event ID 13 • Sysmon Event ID 22 For every rule include: • Title • Description • Log Source • Detection Logic • Investigation Fields • Severity • MITRE ATT&CK Mapping Recommend methods to reduce false positives.

Sigma Detection Rules for SOC Monitoring

The following Sigma rules are designed to detect the requested Windows and Sysmon events. They are generic examples that can be customized for your environment.

1. Windows Event ID 4625 – Multiple Failed Login Attempts

title: Multiple Failed Login Attempts Against Privileged Account

id: 1a8f9e20-4625-001

status: experimental

description: >

Detects failed logon attempts against privileged accounts that may indicate brute-force or password guessing activity.

references:

- <https://attack.mitre.org/techniques/T1110/>

author: SOC Analyst

date: 2026-07-03

logsource:

product: windows

service: security

detection:

selection:

EventID: 4625

TargetUserName:

- Administrator
- admin

condition: selection

fields:

- TargetUserName
- IpAddress
- WorkstationName
- LogonType
- FailureReason
- Status
- SubStatus

falsepositives:

- User entering incorrect password
- Expired credentials

- Service account authentication issues

level: medium

tags:

- attack.credential-access
- attack.brute-force
- attack.T1110

Investigation Fields

- TargetUserName
- Source IP Address
- Logon Type
- Failure Reason
- Workstation Name
- Status/SubStatus codes
- Timestamp

Severity

Medium (Increase to High if repeated attempts occur within a short period.)

MITRE ATT&CK Mapping

- **T1110 – Brute Force**

Reducing False Positives

- Alert only after a threshold (e.g., 5–10 failures in 5 minutes).
- Exclude approved vulnerability scanners.
- Exclude known password synchronization systems.
- Baseline normal failed login behavior.

2. Sysmon Event ID 1 – Encoded PowerShell Execution

title: PowerShell Executed with Encoded Command

id: 7b3c2e91-sysmon-001

status: stable

description: >

Detects PowerShell launched using the EncodedCommand parameter, commonly associated with obfuscated or malicious execution.

references:

- <https://attack.mitre.org/techniques/T1059/001/>

author: SOC Analyst

date: 2026-07-03

logsource:

product: windows

category: process_creation

detection:

selection:

EventID: 1

Image|endswith: '\powershell.exe'

command:

CommandLine|contains:

- '-EncodedCommand'

- '-enc'

condition: selection and command

fields:

- Image
- ParentImage
- CommandLine
- User
- ProcessGuid
- ParentProcessGuid
- Hashes

falsepositives:

- Administrative automation
- Configuration management tools
- Backup software

level: high

tags:

- attack.execution
- attack.T1059.001
- attack.defense-evasion
- attack.T1027

Investigation Fields

- CommandLine
- Parent Process
- User
- Process GUID
- Hashes
- Parent Process GUID

Severity

High

MITRE ATT&CK Mapping

- T1059.001 – PowerShell
- T1027 – Obfuscated Files or Information

Reducing False Positives

- Whitelist approved PowerShell automation.
- Exclude trusted administration platforms.
- Alert only for unexpected users or hosts.
- Decode Base64 commands before triage.

3. Sysmon Event ID 13 – Registry Run Key Modification

title: Registry Run Key Modification

id: e94a5f10-sysmon-013

status: stable

description: >

Detects modifications to Windows Run registry keys that may indicate persistence.

references:

- <https://attack.mitre.org/techniques/T1547/001/>

author: SOC Analyst

date: 2026-07-03

logsource:

product: windows

category: registry_set

detection:

selection:

EventID: 13

target:

TargetObject|contains:

- '\Software\Microsoft\Windows\CurrentVersion\Run'
- '\Software\Microsoft\Windows\CurrentVersion\RunOnce'

condition: selection and target

fields:

- Image
- TargetObject

- Details
- User
- ProcessGuid

falsepositives:

- Software installation
- Operating system updates
- Enterprise software deployment

level: high

tags:

- attack.persistence
- attack.T1547.001

Investigation Fields

- Registry Path
- Process Image
- Modified Value
- User
- Process GUID

Severity

High

MITRE ATT&CK Mapping

- **T1547.001 – Registry Run Keys / Startup Folder**

Reducing False Positives

- Exclude approved software installers.
- Baseline known startup entries.
- Alert only on new or modified values.
- Correlate with recent process creation events.

4. Sysmon Event ID 22 – Suspicious DNS Queries

title: Suspicious DNS Queries from Process

id: 8d91c4b2-sysmon-022

status: experimental

description: >

Detects repeated DNS queries that may indicate malware beaconing, command-and-control, or domain generation algorithm (DGA) activity.

references:

- <https://attack.mitre.org/techniques/T1071/004/>

author: SOC Analyst

date: 2026-07-03

logsource:

product: windows

category: dns_query

detection:

selection:

EventID: 22

suspicious_process:

Image|endswith:

- '\powershell.exe'
- '\cmd.exe'
- '\wscript.exe'
- '\cscript.exe'

condition: selection and suspicious_process

fields:

- Image
- QueryName
- QueryStatus
- QueryResults
- User
- ProcessGuid

falsepositives:

- Security tools
- Software updates
- Web browsers
- Cloud management agents

level: medium

tags:

- attack.command-and-control
- attack.T1071.004

Investigation Fields

- Query Name
- Query Results
- Process Name
- User
- Process GUID

- DNS Response

Severity

Medium (Increase to High if correlated with encoded PowerShell, malware alerts, or repeated beaconing.)

MITRE ATT&CK Mapping

- **T1071.004 – Application Layer Protocol: DNS**

Reducing False Positives

- Whitelist trusted update and security service domains.
- Alert on newly observed or low-reputation domains.
- Use frequency thresholds to identify repeated beaconing.
- Correlate DNS activity with suspicious process creation and network connections.

Summary Table

Sigma Rule	Windows/Sysmon Event	Severity	MITRE ATT&CK
Multiple Failed Login Attempts	Windows Event ID 4625	Medium	T1110 – Brute Force
Encoded PowerShell Execution	Sysmon Event ID 1	High	T1059.001 – PowerShell, T1027 – Obfuscated Files or Information
Registry Run Key Modification	Sysmon Event ID 13	High	T1547.001 – Registry Run Keys / Startup Folder
Suspicious DNS Queries	Sysmon Event ID 22	Medium	T1071.004 – Application Layer Protocol: DNS

General Recommendations to Reduce False Positives

- Build environment-specific baselines for authentication, PowerShell usage, registry changes, and DNS activity.
- Maintain allowlists for trusted administrative tools, software deployment systems, security products, and update services.
- Use threshold-based detections (for example, multiple failed logons or repeated DNS queries within a defined time window) instead of single-event alerts.
- Correlate related events (e.g., failed logons → successful logon → encoded PowerShell → outbound network activity) to increase detection confidence.
- Incorporate threat intelligence to prioritize alerts involving known malicious IP addresses, domains, or file hashes.

- Regularly review and tune Sigma rules based on incident investigations and changes in the environment to minimize unnecessary alerts while preserving detection coverage.